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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: ENGINEERING CAPSTONE PROJECT – 1(23IE4053R/23IE4053A)
A.Y. 2026 - 2027

TITLE OF THE PROJECT	Comparative Study of Recommendation Systems Using Machine Learning	
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Abstract

Recommendation systems are software systems that help users find items or content that may be useful or interesting to them. They are widely used in applications such as online shopping, movie and music platforms, social media, and online learning. These systems study information such as user preferences, previous activities, ratings, and item details to provide personalized recommendations.

This project presents a comparative study of recommendation systems using machine learning. The main aim is to understand how different recommendation techniques work and to compare their performance. The study considers three commonly used approaches: collaborative filtering, content-based filtering, and hybrid recommendation. Collaborative filtering recommends items by finding similarities between users and their interactions. Content-based filtering recommends items that are similar to items a user has already liked or selected. Hybrid recommendation combines more than one approach to improve the quality of recommendations.

For the study, a suitable recommendation dataset is used and the required data is preprocessed before applying the machine learning methods. The models are evaluated using suitable performance measures such as accuracy, precision, recall, or other recommendation quality measures depending on the dataset and method. The results are then compared to understand which approach performs better under different conditions. The study also considers important challenges such as limited user information, new users or items, and the large number of available items.

The expected outcome of this project is a clear comparison of different recommendation methods and an understanding of their advantages and limitations. The study can help identify which recommendation approach is more suitable for a particular application. Overall, this project demonstrates how machine learning can be used to provide more relevant, personalized, and useful recommendations to users.

Signature of the Guide HOD

Batch 06

Comparative Study of Recommendation Systems Using Machine Learning

Domain

- Machine Learning
- Recommendation Systems

Dataset

1. MovieLens 100K / 1M (user-movie ratings, genres, timestamps)
   https://grouplens.org/datasets/movielens/
2. Amazon Reviews (real-world e-commerce reviews with product metadata)
   https://www.kaggle.com/datasets/snap/amazon-fine-food-reviews
3. Book-Crossing (book ratings with demographic info)
   https://www.kaggle.com/datasets/saurabhbagchi/book-crossing-dataset

Abstract

Recommendation systems have become an essential component of modern digital platforms by helping users discover relevant products, movies, books, and other items based on their preferences and historical interactions. Different recommendation techniques exhibit varying levels of performance depending on the characteristics of the dataset and the recommendation domain. This project presents a comparative study of multiple recommendation system approaches using benchmark datasets, including MovieLens 100K/1M, Amazon Product Reviews, and the Book-Crossing Dataset. The study evaluates the effectiveness of Popularity-Based Recommendation, Content-Based Filtering, Collaborative Filtering (User-Based and Item-Based), Matrix Factorization, and Machine Learning-based recommendation models.

The project begins with data collection, followed by data preprocessing, which includes handling missing values, removing duplicates, encoding categorical variables, and normalizing the data. Exploratory Data Analysis (EDA) is then performed to understand user behaviour, item popularity, rating distributions, and feature relationships. Subsequently, different recommendation algorithms are implemented and evaluated using appropriate performance metrics such as RMSE, MAE, Precision, Recall, F1-Score, and NDCG. The performance of each recommendation model is compared to identify its strengths, limitations, and suitability for different application domains.

The findings of this study provide valuable insights into selecting the most effective recommendation technique based on dataset characteristics, computational complexity, and recommendation accuracy. The project also demonstrates how machine learning techniques can enhance personalized recommendations, thereby improving user satisfaction and decision-making across various online platforms such as e-commerce, streaming services, and digital libraries.

Data Analytics (EDA)

Raw Dataset

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Dataset Overview

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Data Quality Assessment

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Descriptive Statistics

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Missing Values

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Duplicate Records

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Outlier Detection

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Univariate Analysis

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Bivariate Analysis

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Correlation Analysis

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Feature Engineering

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Feature Selection

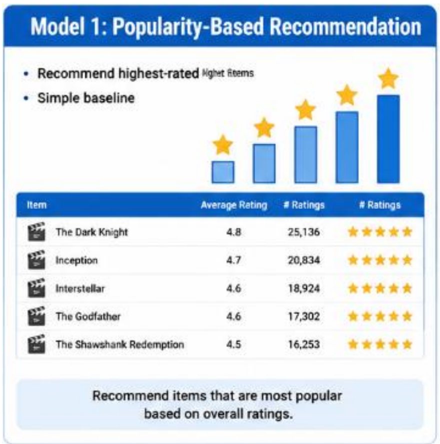
Data Preprocessing

- Handle missing values
- Remove duplicates
- Encode User IDs
- Encode Item IDs
- Normalize ratings

- Feature scaling

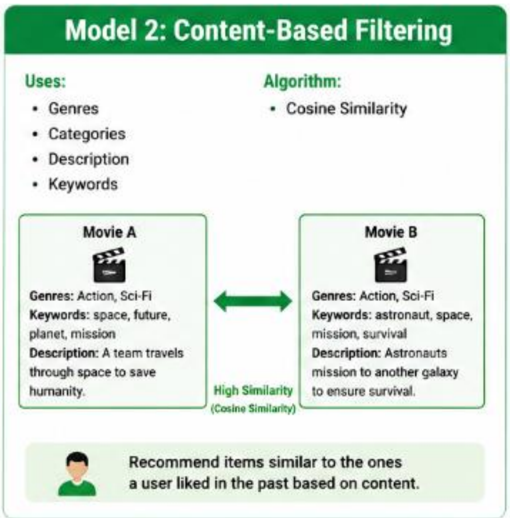
Recommendation Algorithms

Compare different recommendation techniques.



Model 1: Popularity-Based Recommendation

- Recommend highest-rated items
- Simple baseline



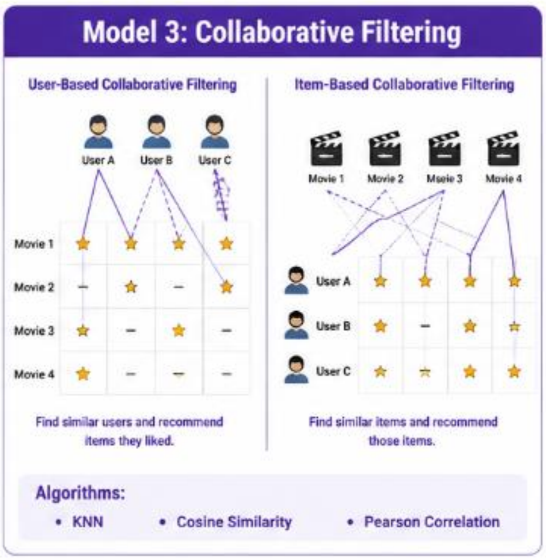
Model 2: Content-Based Filtering

Uses:

- Genres
- Categories
- Description
- Keywords

Algorithms:

- Cosine Similarity



Model 3: Collaborative Filtering

User-Based Collaborative Filtering

Item-Based Collaborative Filtering

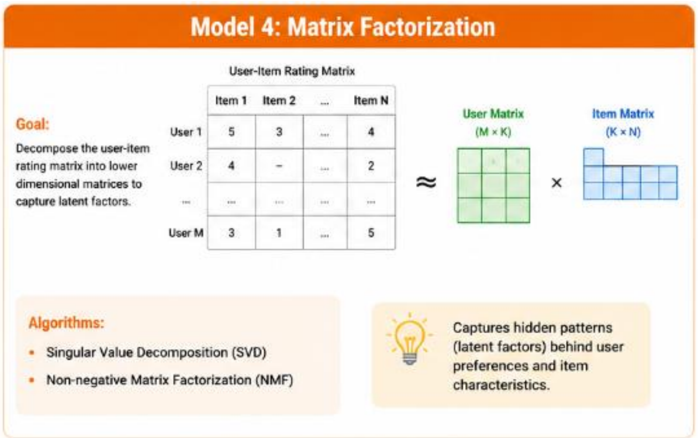
Algorithms:

- KNN
- Cosine Similarity
- Pearson Correlation

Model 4: Matrix Factorization

Algorithms:

- Singular Value Decomposition (SVD)
- Non-negative Matrix Factorization (NMF)

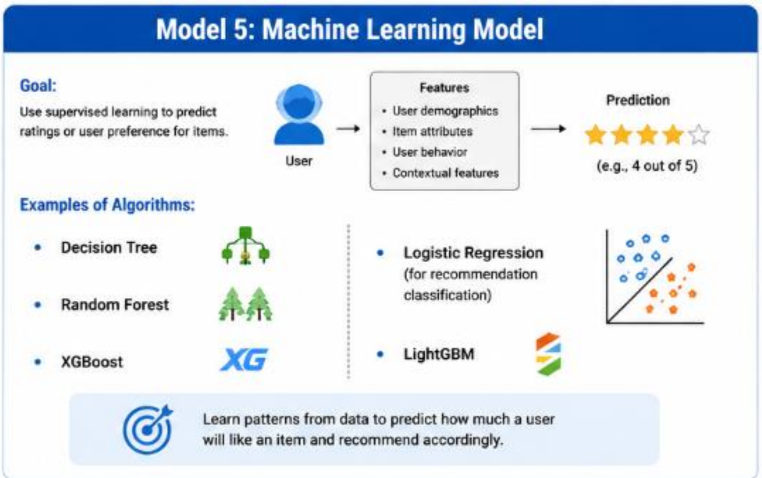


Model 5: Machine Learning Model

Use supervised learning to predict ratings.

Examples:

- Decision Tree
- Random Forest
- XGBoost
- Logistic Regression
- LightGBM



Evaluation Metrics

Recommendation systems are commonly evaluated using:

Rating Prediction

- RMSE
- MAE

Recommendation Quality

- Precision
- Recall
- F1-Score
- MAP (Mean Average Precision)

- Hit Ratio

Comparative Analysis

- Accuracy
- RMSE
- MAE
- Precision
- Recall
- F1-Score
- NDCG

Research Pipeline

Benchmark Dataset

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Data Cleaning

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EDA

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Feature Engineering

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Feature Selection

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Train-Test Split

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Recommendation Models

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Performance Evaluation

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Visualization

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Comparative Analysis

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Research Conclusions

Expected Outcome

- Compare traditional recommendation methods with machine learning approaches.
- Identify the best-performing algorithm based on recommendation quality and prediction accuracy.
- Analyze trade-offs between accuracy, computational complexity, and scalability.
- Provide insights into which recommendation technique is most suitable for real-world applications such as movie, e-commerce, or book recommendation systems.